

Question				Marking details		Marks available																	
						AO1	AO2	AO3	Total	Maths	Prac												
4	(a)			<table><tr><td>Umbilical artery</td><td>Umbilical vein</td></tr><tr><td>Less oxygen</td><td>More oxygen</td></tr><tr><td>More CO₂</td><td>Less CO₂</td></tr><tr><td>More urea</td><td>Less urea</td></tr><tr><td>Less nutrients/named</td><td>More nutrients/named</td></tr><tr><td>Less antibodies</td><td>More antibodies</td></tr></table>		Umbilical artery	Umbilical vein	Less oxygen	More oxygen	More CO ₂	Less CO ₂	More urea	Less urea	Less nutrients/named	More nutrients/named	Less antibodies	More antibodies	2			2		
				Umbilical artery	Umbilical vein																		
				Less oxygen	More oxygen																		
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				More urea	Less urea																		
				Less nutrients/named	More nutrients/named																		
				Less antibodies	More antibodies																		
	(b)			Any two (x1) from: (Barrier) {against hormones/toxic substances/microorganisms/cells/mothers rhesus group/mothers antigens } (1) Protection against differences in pressure (1) Protection against mother's {immune system/antibodies} (1)		2			2														
	(c)	(i)		Maintain {concentration/diffusion} gradients/prevents equilibrium being reached(1)																			
		(ii)		(Pressure difference) forces materials through (capillaries) (1)																			
		(iii)		Large (surface) area for exchange.(1)																			
	(d)	(i)		GGC GTA ATT CCC			1		1														
		(ii)		different nucleotide sequence at each end/one primer for each strand of DNA (1).				1	1														
		(iii)		Enables {specific/the gene} to be {amplified/to be copied/to be replicated} (1)				1	1														
		(iv)		Does not matter how {many cycles in PCR/initial quantity/number of copies made}(1) Ratio will be same (1)				2	2														